Spatial-Frequency Synergistic Graph-Aware Bidirectional Optimization Network for Remote Sensing Change Detection

Yankang Liu, Anyu Du, Shuli Cheng, *Member, IEEE*, and Chunpeng Wu

***Abstract*—Remote sensing change detection (RSCD) is a fundamental approach for surface monitoring, disaster assessment, and urban evolution analysis. Accurate detection of changes in remote sensing images remains challenging due to factors such as illumination variations and tree occlusions. Many existing approaches exhibit weakened discriminative performance when utilizing shallow detail features, since shadowing and occlusion effects can conceal important change cues. To address these challenges, we propose a spatial-frequency synergistic graph-aware bidirectional optimization network (SFGBO), which improves change discrimination and feature representation from two complementary perspectives: difference feature identification and effective shallow feature fusion. Specifically, we design a spatial-frequency synergistic dynamic attention module (SF-DAM) that jointly models spatial structures and frequency responses on a coarse difference map, enabling the suppression of local pseudo-changes while emphasizing true change regions. Furthermore, we introduce a graph-based bidirectional dynamic modeling module (GBDM), which leverages graph perception and deformable convolutions to enhance the modeling of nonrigid changes and multiscale boundaries. A bidirectional optimization strategy is then employed to adaptively integrate shallow details with high-level semantics, thereby improving the network's ability to capture complex structural changes. Comprehensive experiments on three benchmark datasets—BCDD, LEVIR-change detection (CD), and SYSU-CD—validate that SFGBO achieves superior performance compared with recent state-of-the-art approaches in both visual quality and numerical metrics. The code is available at https://github.com/lyk-2891/SFGBO**

***Index Terms*—Change detection (CD), graph-based bidirectional dynamic modeling module (GBDM), spatial-frequency synergistic dynamic attention module (SF-DAM).**

I. Introduction

REMOTE sensing change detection (RSCD) constitutes a core component of geospatial data analysis, focusing on the identification and interpretation of land-cover variations within identical geographic regions observed at multiple time intervals. Given its growing significance in tracking environmental evolution and human-induced transformations, RSCD has found extensive use in domains including urban development planning [1], agricultural land supervision [2], and water resource management [3].

Conventional change detection (CD) techniques are generally divided into three primary groups: arithmetic-based methods, image transformation-based methods, and classification-based methods [4], [5], [6], [7]. Arithmetic-based methods usually compute difference maps through pixel-level comparisons of bitemporal images and subsequently employ thresholding or statistical decision rules to locate regions of change. Representative approaches include image differencing [8], image ratioing [9], and image regression analysis [10]. Image transformation-based methods project the images into a feature space to extract more discriminative change cues. Typical examples are principal component analysis (PCA) [11], multivariate alteration detection (MAD) [12], and change vector analysis (CVA) [13]. The rise of machine learning has driven a growing adoption of classification-oriented frameworks, such as decision trees [14], random forests [15], and support vector machines (SVMs) [16]. However, these traditional methods often rely heavily on expert knowledge and are sensitive to the quality of remote sensing imagery, which limits their robustness and generalizability. Benefiting from powerful feature representation and nonlinear modeling abilities, deep neural networks have gradually become the dominant paradigm for RSCD. Existing deep learning approaches have evolved from convolutional neural networks (CNNs) [17], [18], [19], [20], to Transformer-based models [21], [22], [23], and more recently to Mamba-based architectures [24], [25], continually advancing RSCD through improvements in local feature extraction, global context modeling, and efficient long-range dependency learning. Although significant progress has been made in existing research, issues such as insufficient detection of subtle changes and limited ability to suppress spurious changes in complex remote sensing scenarios still persist.

As illustrated in Fig. 1(a), the same building exhibits spectral differences under varying illumination conditions, resulting in pseudo-change issues. In Fig. 1(b), occlusion by trees reduces feature discriminability, leading to missed detections. Fig. 1(c) shows edge misdetections caused by shadows. We consider these issues as manifestations of

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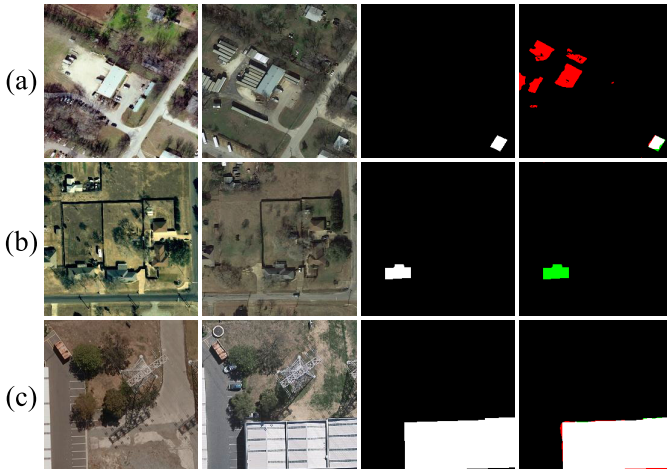


Fig. 1. Various unrelated changes affect the CD results. (a) Illumination intensity. (b) Tree occlusion. (c) Shadows. The first and second columns correspond to the images at Time1 (T1) and Time2 (T2). The third and fourth columns show the ground truth and the possible change prediction map. Changed objects are highlighted in white, while unchanged objects are shown in black. Red regions indicate false detections, and green regions denote missed detections.

insufficient model discriminability. Existing studies indicate that effectively leveraging shallow features can mitigate the degradation of deep features in fine-detail representation, thereby enhancing the model's ability to detect subtle changes. However, mainstream difference modeling methods can be categorized into three groups: absolute subtraction, feature concatenation, and cross-attention. The first two approaches are structurally simple and computationally efficient, but their ability to capture complex change details is limited, often resulting in missed detections of small changes. In contrast, methods that combine cross-attention with multiscale mechanisms [26], [27], [28] can enhance the modeling of shallow difference features and improve the responsiveness to change regions, but they are typically associated with high computational cost. Some studies [29], [30], [31], [32], [33], [34] have attempted to introduce frequency-domain modeling into RSCD as a complement to spatial-domain difference modeling. This enhances the model's ability to perceive subtle changes to some extent. However, most of them regarded the frequency domain as a separate feature enhancement space, which somewhat constrained the guidance of spectral response to the spatial representation of disparity perception. Moreover, the effective fusion of shallow features with deep semantic features is crucial for enhancing overall CD performance. Existing methods typically introduce shallow features into the decoding branch through skip connections to recover spatial details. However, due to the significant differences in scale and distribution between shallow and deep features, simple concatenation or weighted fusion often leads to feature conflicts, thereby degrading the decoding quality. Although some studies have leveraged attention mechanisms to partially alleviate this issue, most of them overlook the collaborative relationships between shallow and deep features. As a result, shallow information is introduced indiscriminately, leading to inefficient utilization of valuable details.

Motivated by the above challenges, we propose a novel CD network, termed spatial-frequency synergistic graph-aware bidirectional optimization network (SFGBO). The proposed network introduces a spatial-frequency synergistic dynamic attention module (SF-DAM) to adaptively highlight local structural and textural differences within fine-grained change features. In addition, a graph-based bidirectional dynamic modeling module (GBDM) is incorporated to achieve complementary fusion between shallow details and high-level semantics, thereby simultaneously enhancing the sensitivity to subtle changes and suppressing pseudo-changes. The main contributions of this work can be summarized as follows.

1) We propose a novel CD network, SFGBO. Extensive quantitative and qualitative experiments on three public datasets confirm that SFGBO delivers competitive results, demonstrating the effectiveness of our method.
2) We propose an SF-DAM, which jointly models spatial structures and frequency response information. This module is designed to suppress irrelevant noise while highlighting true change regions, thereby enhancing the robustness and discriminative capability of the model in complex remote sensing scenarios.
3) We design a GBDM, which employs a dynamic fusion strategy on top of SF-DAM to efficiently leverage shallow feature flows, further enhancing the model's ability to discriminate subtle changes.

The remainder of this article is structured as follows. Section II summarizes the relevant knowledge used in this article. Section III explains the core work of this article in detail. Section IV quantitatively and qualitatively analyzes the effectiveness of our model. Section V summarizes the work of this article.

II. Related Work

A. Difference Feature Enhancement Strategy for RSCD

In RSCD, different features are crucial for distinguishing changed regions from the background, and their quality directly affects detection accuracy and robustness. However, due to radiometric differences, registration errors, illumination variations, and complex textures across temporal images, raw difference maps often contain substantial noise and pseudo-change signals. Difference feature enhancement aims to suppress background interference while preserving true changes, thereby improving the separability and representational power of change regions in the feature space. This aspect becomes crucial under complex conditions, as it effectively strengthens the model's ability to generalize across diverse situations. For instance, Li et al. [35] proposed a multiscale difference enhancement module in MDFENet, which constructs a multiscale architecture combining CNNs and Transformers. This module fuses semantic cues from high-level and low-level tokens to accurately localize change objects. Song et al. [36] proposed a contextual difference enhancement network, introducing a CDEM module to enhance difference features. The module extracts detailed positional information of different features using spatial attention and applies the resulting attention weights to features from both temporal

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images. Lei et al. [37] designed a difference enhancement module that leverages channel attention (CAM) and residual connections to suppress irrelevant noise, thereby capturing difference features between the foreground and background. Ying et al. [38] designed the GLFF module in AEGL-Net for dual-temporal feature fusion. This module enhances global dependency modeling through global–local attention while integrating shared and context-aware weights to strengthen local feature representation. Overall, although different feature enhancement strategies have continuously evolved in terms of structural design and modeling capability, most existing methods still suffer from limited stability when dealing with high-frequency pseudo-changes, texture interference, and multiscale structural variations. Moreover, they lack an efficient mechanism to jointly integrate spatial and frequency information. To overcome the previously mentioned shortcomings, we propose an SF-DAM, which integrates multireceptive field spatial modeling with frequency-domain enhancement on top of the difference features, effectively improving the separability between true changes and background regions.

B. Feature Fusion Strategies for RSCD

Feature fusion constitutes a key element of RSCD frameworks, intended to integrate spatial and semantic representations across multiple levels extracted from bitemporal remote sensing data. By jointly modeling these complementary representations, the model can more precisely localize change regions while suppressing irrelevant background noise. An effective fusion mechanism improves the model's ability to capture subtle variations and substantially increases robustness under complex and dynamic imaging conditions. To enhance the utilization of edge information, Jiang and Wu [39] designed an EHM in the decoding stage. This module strengthens the representation of change-region details through direction-aware enhancement and improves contextual interactions among features via channel shuffling and hybrid selection mechanisms, thereby producing more robust detection results. To mitigate insufficient interaction between bitemporal features and information loss across layers, Xie et al. [40] proposed a dual-complementary attention module. The module employs dense skip connections to supplement semantic information from different feature levels and subsequently captures both interlayer and intralayer dependencies to obtain discriminative feature maps. Additionally, Liu et al. [41] designed a dual-pyramid architecture, where the S-FIPN leverages the global semantic information from top-level low-resolution features to guide lower-level features toward key change regions, enabling fine-grained feature extraction. To suppress pseudo-changes caused by imaging style variations, Han et al. [42] introduced a bitemporal feature fusion module guided by change priors with cross-attention in the decoding stage. This module adaptively enhances change information, improving the model's sensitivity to subtle changes. Despite the promising results achieved by the aforementioned methods, most rely on static fusion strategies. Such indiscriminate fusion can induce semantic conflicts or feature redundancy, thereby weakening the discriminative power of the decoding stage and increasing the likelihood of misclassification, particularly in complex background scenarios. To alleviate these problems, a GBDM is introduced, enabling refined joint modeling between shallow details and semantic-level representations throughout the fusion process. This approach effectively mitigates semantic conflicts and feature redundancy, thereby enhancing both the accuracy of change-region identification and the precision of boundary delineation.

C. Frequency Domain Study on RSCD

In recent years, frequency-domain modeling has received wide attention in RSCD. Since the frequency representation can portray the global structure and local change features of the image from the spectral domain perspective, the related methods usually take the frequency domain information as an important complement to the spatial features. Ma et al. [29] introduced a dual-domain learning strategy in DDLNet, which combined the spatial representation with the frequency-aware features, and mapped the bitemporal inputs to the frequency domain by discrete cosine transform to enhance the feature responses related to the change regions. Zhang et al. [30] proposed the global awareness operator (GAO) in GCRNet, which utilizes the intrinsic connection between convolution and Fourier transform to achieve effective modeling of global contextual information with low computational overhead. Liu et al. [31] constructed a multidomain disparity mining network in MDENet to alleviate the inconsistency of image styles caused by seasonal changes and illumination differences, and through MDCM jointly modeled the frequency domain features and spatial domain target features in different time domains to achieve the synergistic portrayal of global and local change information. In addition, Jie et al. [32] designed SDA and WFDA in SFDA-OCDet to enhance the difference features of changing objects in the spatial and frequency domains, respectively, and Song et al. [33] introduced wavelet transform in WaveHFG to extract the high-frequency information to enhance the robustness of the heterogeneous remote sensing image representations in terms of similarity and directional consistency. Although the above methods have made some progress in introducing frequency domain information, most of the frequency-based RSCD methods still regard the frequency domain as an independent feature space, and the spatial domain and frequency domain features are usually augmented in relative isolation, and the synergistic modeling between them is still limited. This somewhat constrains the usefulness of spectral response in guiding the spatial representation of disparity perception.

III. Methods

A detailed overview of the proposed SFGBO is presented in this section. Section III-A introduces the overall network architecture, while Sections III-B and III-C elaborate on two key components: the SF-DAM and the GBDM. The loss function and its role in network optimization are detailed in Section III-D.

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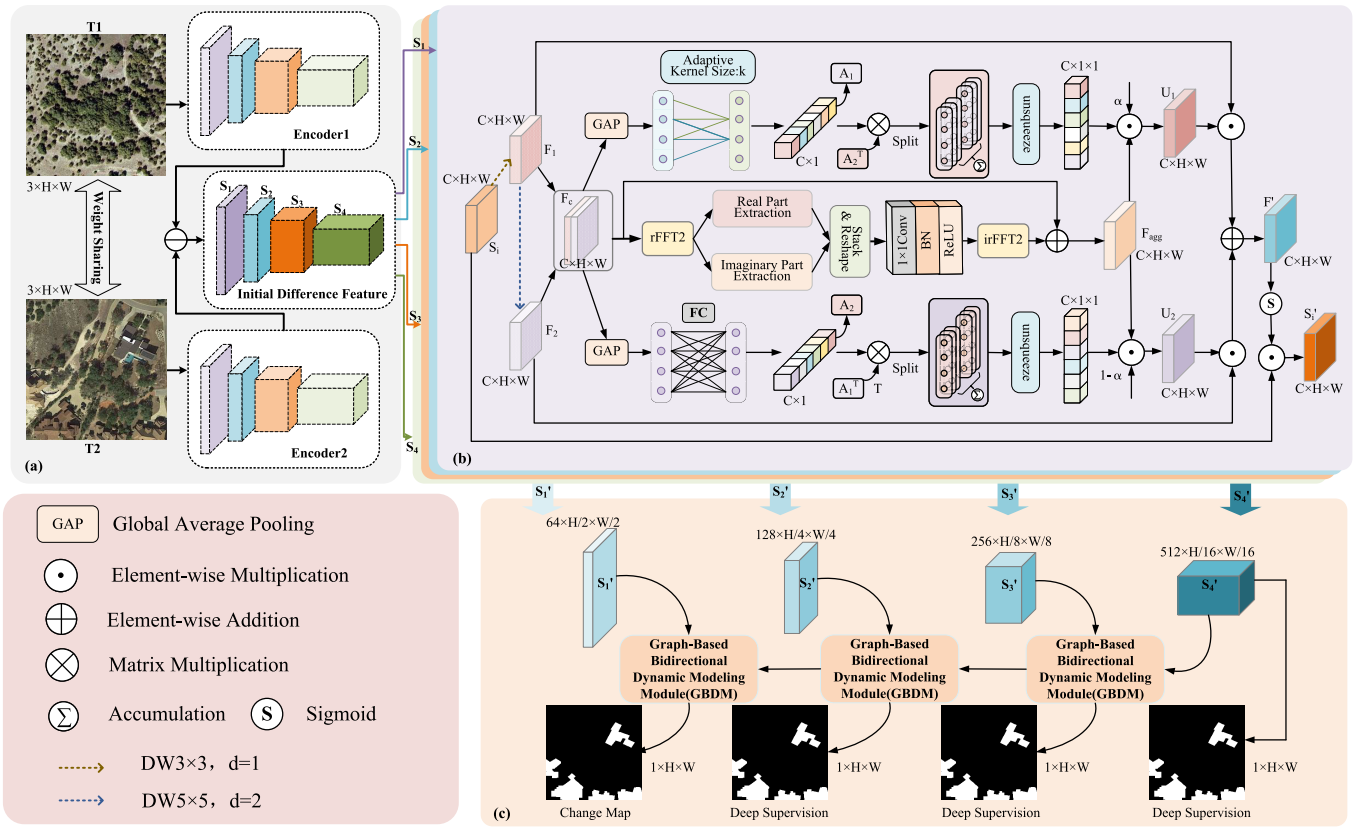


Fig. 2. Overall architecture of the proposed SFGBO. The framework comprises three key parts: (a) feature extraction network that encodes bitemporal inputs and generates an initial difference feature. (b) SF-DAM, which integrates spatial and frequency cues to adaptively emphasize change-related features while suppressing background noise. (c) Multilevel decoder enhanced by the GBDM progressively refines predictions by capturing fine-grained interfeature details and modeling bidirectional dependencies.

A. Architecture Overview

To enhance the sensitivity to subtle changes in RSCD tasks, our approach focuses on two aspects: difference feature enhancement and effective fusion of shallow features. By jointly addressing these aspects, we improve both the accuracy and robustness of CD. As illustrated in Fig. 2, the model takes a remote sensing image pair at T_1 and T_2 as input, which is then passed through a VGG16 backbone with shared weights. Multiscale initial difference features S_1, S_2, S_3, and S_4 are obtained layer by layer using absolute difference. Subsequently, the SF-DAM module is introduced to adaptively enhance the discriminative capability between change regions and the background by leveraging a coordinated strategy across spatial and frequency domains, with a particular focus on highlighting subtle texture and structural differences of small-scale targets. To further enhance the model's ability to perceive and model the structural information of multiscale change regions, we introduce the GBDM in the decoding stage. This module dynamically selects and fuses shallow difference features with deep semantic representations through a multibranch architecture to achieve complementary information integration. Specifically, GBDM leverages an enhanced feature-driven graph modeling mechanism to strengthen the representation of spatial topological relationships and edge details. Moreover, a bidirectional optimization strategy is employed to enable efficient collaboration and dynamic fusion among branches. Finally, multiscale auxiliary supervision is applied to optimize the outputs of each GBDM layer, resulting in a refined, high-resolution pixel-level change map.

B. Spatial-Frequency Synergistic Dynamic Attention Module

Due to illumination variations, shadow occlusions, and other environmental factors, the true change details in difference features are often overwhelmed by irrelevant noise, which weakens the ability to detect subtle target changes. To address this problem, we propose the SF-DAM. Building upon the basic difference map, SF-DAM introduces a collaborative modeling strategy that jointly leverages the spatial and frequency domains to adaptively enhance the separability between change regions and background areas.

As illustrated in Fig. 2(b), the architecture of SF-DAM takes four groups of multiscale features, denoted as S_1, S_2, S_3, and S_4. These features are sequentially processed by two depthwise convolution operations with different dilation rates to capture spatial structural information under varying receptive fields. Subsequently, the resulting features generated by different kernels are passed through a 1×1 convolution layer to integrate channel information and unify feature dimensions. This process can be formulated as

$$F_1 = \mathrm{Conv}_{1\times1}\left(\mathrm{DW}_{3\times3}\left(S_i\right)\right), \quad i \in \{1,2,3,4\} \tag{1}$$

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$$F_2 = \mathrm{Conv}_{1\times1}\left(\mathrm{DW}_{5\times5}\left(\mathrm{DW}_{3\times3}\left(S_i\right)\right)\right), \quad i \in \{1,2,3,4\} \tag{2}$$

$$F_c = \mathrm{Cat}\left(F_1, F_2\right) \tag{3}$$

where DW3 × 3(·) and DW 5 × 5 (·) denote depthwise separable convolutions with kernel sizes 3 × 3 and 5 × 5, respectively, and use dilation rates of 1 and 2. $\mathrm{Conv}_{1\times1}(\cdot)$ refers to a 1 × 1 convolution with batch normalization, and Cat(·) indicates the channel-wise concatenation.

To further explore the structural stability and spectral responses of change regions, we perform a frequency-domain enhancement on the fused feature $F_c \in \mathbb{R}^{C\times H\times W}$. Inspired by the Fourier unit in fast Fourier convolution (FFC) [43], we adopt a real-valued Fourier transform formulation to extract frequency-domain responses from difference-aware features, which are subsequently used to guide attention-based feature modulation. Specifically, we implement this process using rFFT2/irFFT2. For a real-valued input, rFFT2 exploits the Hermitian symmetry of the full Fourier spectrum and returns a nonredundant one-sided frequency representation

$$\hat{F}_c = \mathrm{rFFT2}\left(F_c\right) \in \mathbb{C}^{C\times H\times(W/2+1)}. \tag{4}$$

To enable real-valued convolution while preserving the correspondence between real and imaginary components, we explicitly separate them and organize them as paired components along an additional dimension

$$F_{ri} = \mathrm{Stack}\left(\Re\left(\hat{F}_c\right), \Im\left(\hat{F}_c\right)\right) \in \mathbb{R}^{C\times2\times H\times(W/2+1)} \tag{5}$$

where $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of the complex spectrum, respectively. The second dimension of F_{ri} indexes the paired real and imaginary components of each frequency coefficient. Thus, for each channel c and frequency location (u, v), $F_{ri}[c, 0, u, v]$ and $F_{ri}[c, 1, u, v]$ correspond to the real and imaginary components of the same frequency element, respectively. The tensor is then reshaped into the channel dimension to obtain a real-valued representation

$$F_{cat} = \mathrm{Reshape}\left(F_{ri}\right) \in \mathbb{R}^{2C\times H\times(W/2+1)}. \tag{6}$$

This representation enables the subsequent real-valued convolution to jointly model real-imaginary correlations and cross-channel dependencies in the frequency domain. Accordingly, a real-valued 1 × 1 convolution followed by batch normalization and ReLU activation is applied

$$F_{proc} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(F_{cat}\right)\right)\right). \tag{7}$$

After transformation, the processed tensor is reshaped back to restore the paired real-imaginary structure

$$\tilde{F}_{ri} = \mathrm{Reshape}\left(F_{proc}\right) \in \mathbb{R}^{C\times2\times H\times(W/2+1)}. \tag{8}$$

The paired components are then recombined into a complex-valued one-sided spectrum

$$\tilde{F}_c = \tilde{F}_{ri}[:, 0, :, :] + j \cdot \tilde{F}_{ri}[:, 1, :, :] \tag{9}$$

where j denotes the imaginary unit. Finally, the enhanced frequency representation is transformed back to the spatial domain using irFFT2 with the original spatial size

$$F_{freq} = \mathrm{irFFT2}\left(\tilde{F}_c, s = (H, W)\right) \tag{10}$$

where $s = (H, W)$ specifies the original spatial resolution for inverse transformation. In practice, irFFT2 reconstructs the spatial signal under the real-FFT convention by internally handling the conjugate symmetry of the one-sided spectrum and returns a real-valued tensor, which is directly used in subsequent network layers. The frequency-enhanced feature is then added residually to the fused feature F_c, yielding the spatiofrequency aggregated representation F_{agg}. This mechanism enhances the model's responsiveness to genuine changes under complex backgrounds and improves the collaborative modeling of spatial and frequency information. This process can be formulated as

$$F_{\mathrm{agg}} = F_c + F_{freq}. \tag{11}$$

The aggregated feature F_{agg} is subsequently treated as a context-aware modulation factor in the attention generation stage, enabling frequency-domain responses to participate in the adaptive reweighting of spatial features. This encourages the learned representations to emphasize structural consistency rather than raw intensity variations, thereby helping suppress pseudo changes caused by illumination differences and shadow effects. Accordingly, we generate channel-wise attention candidates from F_c and use the spatiofrequency aggregated representation F_{agg} to guide the subsequent feature modulation. We apply global average pooling on the fused feature F_c to extract channel-wise statistics. Two attention candidate vectors, A_1 and A_2, are then generated via a 1-D convolution and a 1 × 1 convolution pathway, respectively. This process can be formulated as

$$A_1 = \mathrm{Conv}_{1\times1}\left(\mathrm{GAP}\left(F_c\right)\right) \tag{12}$$

$$A_2 = \mathrm{Conv}_{1D}\left(\mathrm{GAP}\left(F_c\right)\right) \tag{13}$$

where GAP(·) denotes the global average pooling operation, and $\mathrm{Conv}_{1\mathrm{D}}(\cdot)$ represents the 1-D convolution. The cross-channel response relationships are then computed through an interaction matrix multiplication. This design explicitly models interchannel dependencies without introducing significant computational overhead, thereby enhancing the discriminability of the attention mechanism. Finally, two sets of attention factors, a_1 and a_2, are generated through a sigmoid activation. The overall process can be formulated as

$$a_1 = \alpha \odot \mathrm{Sigmoid}\left(\mathrm{unsqueeze}\left(\sum\left(A_1 \otimes A_2^{\top}\right)\right)\right) \tag{14}$$

$$a_2 = (1-\alpha) \odot \mathrm{Sigmoid}\left(\mathrm{unsqueeze}\left(\sum\left(A_2^{\top} \otimes A_1\right)\right)\right) \tag{15}$$

where ⊗ denotes matrix multiplication, $\sum(\cdot)$ represents element-wise summation over column vectors, unsqueeze(·) indicates the dimension expansion operation, and ⊙ denotes the element-wise multiplication. The learnable parameter α is introduced to balance the contributions of different features. Subsequently, the spatiofrequency aggregated result F_{agg} is utilized as a contextual enhancement factor to further modulate a_1 and a_2, generating two feature modulation branches, U_1 and U_2. This process can be formulated as

$$U_1 = a_1 \odot F_{agg} \tag{16}$$

$$U_2 = a_2 \odot F_{agg}. \tag{17}$$

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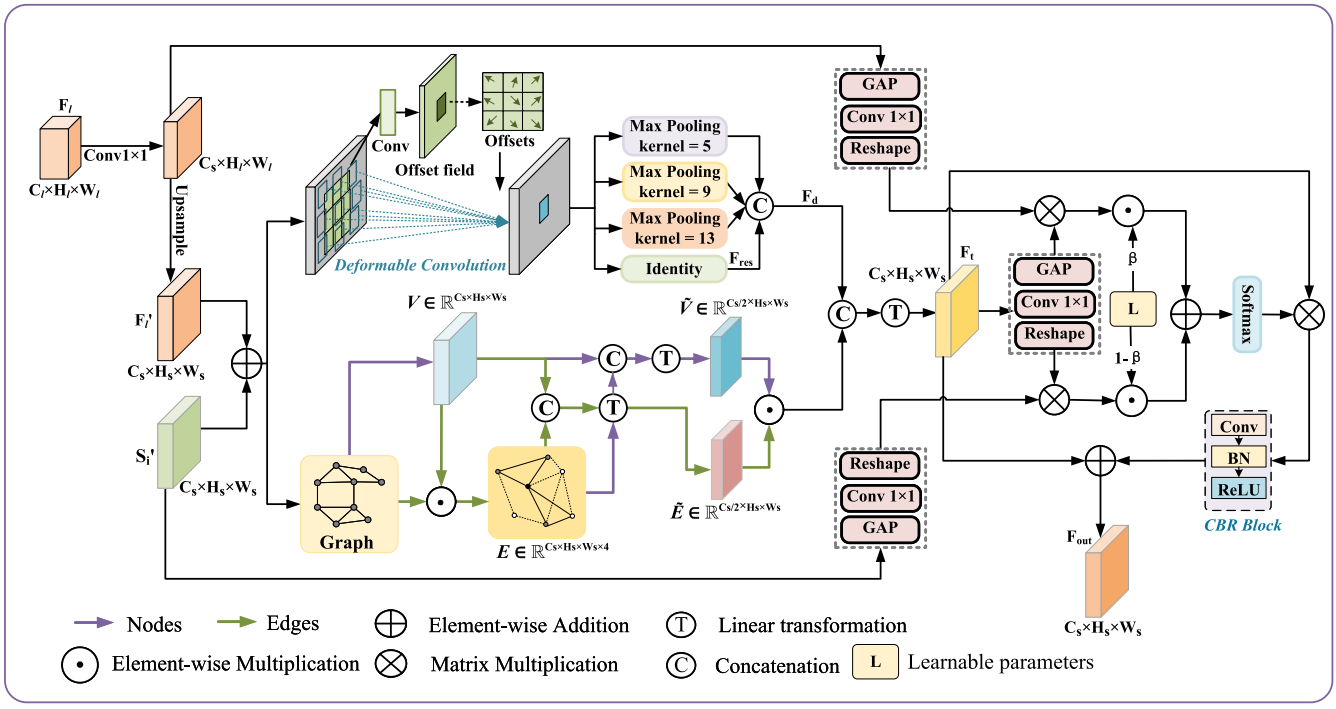


Fig. 3. Graph-based bidirectional dynamic modeling module.

The two feature modulation branches, U_1 and U_2, are element-wise multiplied with F_1 and F_2, respectively. The results are then summed to obtain the fused feature F', thereby enabling adaptive weighting of the two spatial convolution branches. This process can be formulated as

$$F' = U_1 \odot F_1 + U_2 \odot F_2. \tag{18}$$

A 1×1 convolution followed by a Sigmoid activation generates a channel attention map, which is then multiplied element-wise with the input feature S_i to adaptively refine the original difference representation. This process can be formulated as

$$S_i' = S_i \odot \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(F'\right)\right) \tag{19}$$

where S_i' denotes the refined difference feature at the ith scale. In summary, during the difference feature modeling stage, our method effectively preserves detailed information while suppressing irrelevant noise to a certain extent.

C. Graph-Based Bidirectional Dynamic Modeling Module

Skip connections have become a widely adopted strategy to alleviate the loss of fine-grained texture details in shallow layers. However, due to the inherent limitations of shallow features, undesirable noise is inevitably introduced during this process. Although existing methods apply attention mechanisms to refine the fused features, most of them still operate in a static fusion manner, where information from different levels is indiscriminately aggregated and subsequently filtered by attention modules. This strategy not only weakens the discriminative representation of deep features but also increases the modeling burden in the decoding stage, ultimately degrading classification accuracy. To this end, we propose the GBDM, which employs a bidirectional optimization mechanism to enable adaptive interaction and fusion between shallow and high-level features, thereby facilitating the efficient utilization of shallow information.

As illustrated in Fig. 3, the proposed GBDM module takes the low-resolution decoder feature $F_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ from the previous decoding stage as input. First, a 1×1 convolution is applied to adjust its channel dimension to C_s. The transformed feature is then upsampled to match the spatial resolution of the high-resolution shallow feature $S_i' \in \mathbb{R}^{C_s \times H_s \times W_s}$, which is derived from the skip connection branch. This process can be formulated as

$$F_l' = \text{UP}\left(\text{Conv}_{1\times1}\left(F_l\right)\right) \tag{20}$$

where UP$(\cdot)$ denotes the upsampling operation. The upsampled shallow feature is then fused with S_i' through element-wise addition to generate the mixed feature F_{in}, which can be formulated as

$$F_{in} = F_l' + S_i'. \tag{21}$$

In addition, F_{in} is processed through two parallel pathways. On the one hand, to enhance the module's capability of modeling nonrigid changes and local structural deformations, GBDM introduces a deformable convolution path for complementary modeling. Specifically, a 3×3 convolution is first applied to learn the offset field, which guides the sampling locations of the standard feature map F_{in}. Subsequently, a deformable convolution operation is employed to generate the deformation-enhanced feature, which can be formulated as

$$F_{res} = \text{DeformConv}\left(F_{in}, \mathcal{O}\right) \tag{22}$$

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where DeformConv$(\cdot,\cdot)$ represents the deformable convolution operation, and $\mathcal{O}$ denotes the learnable offset field that controls the sampling position shifts in the deformable convolution. By learning these offsets, the deformable convolution can flexibly adjust the sampling locations of the input features, enabling better adaptation to objects of varying shapes. Furthermore, to enhance the model's responsiveness to multiscale structures, three max-pooling operations with kernel sizes of 5, 9, and 13 are applied to F_{res} for multiscale modeling. The resulting features are then concatenated with the original F_{res} to generate the multiscale deformable enhanced feature F_d. This process can be formulated as

$$F_d = \mathrm{Cat}\left(F_{res}, \mathrm{MP}_5\left(F_{res}\right), \mathrm{MP}_9\left(F_{res}\right), \mathrm{MP}_{13}\left(F_{res}\right)\right) \tag{23}$$

where $\mathrm{MP}_k(\cdot)$ represents max-pooling with a kernel of size k. On the other hand, we construct a graph structure to enhance the modeling of local edge information. Specifically, the input feature map F_{in} is first represented as a node set $V \in \mathbb{R}^{C_s \times H_s \times W_s}$. To capture the local structural relationships, the neighboring features of each pixel are gathered from four spatial directions (up, down, left, and right) and concatenated with the original feature. This concatenated representation is then multiplied element-wise with the original feature to generate the structural edge set $E \in \mathbb{R}^{C_s \times H_s \times W_s \times 4}$. This process first defines a horizontal circular shift as follows:

$$\mathrm{Shift}_h^{+1}\left(F_{in}\right)_{b,c,h,w} = \left(F_{in}\right)_{b,c,h,(w-1) \bmod W} \tag{24}$$

$$\mathrm{Shift}_h^{-1}\left(F_{in}\right)_{b,c,h,w} = \left(F_{in}\right)_{b,c,h,(w+1) \bmod W}. \tag{25}$$

The vertical circular shift is defined as follows:

$$\mathrm{Shift}_v^{+1}\left(F_{in}\right)_{b,c,h,w} = \left(F_{in}\right)_{b,c,(h-1) \bmod H,w} \tag{26}$$

$$\mathrm{Shift}_v^{-1}\left(F_{in}\right)_{b,c,h,w} = \left(F_{in}\right)_{b,c,(h+1) \bmod H,w} \tag{27}$$

where mod denotes the modulo operation, which ensures circular boundary conditions such that pixels shifted out from one edge reappear on the opposite edge. Accordingly, the edge features can be formulated as

$$\begin{aligned} E = \mathrm{Stack}\Big[&\mathrm{Shift}_h^{-1}\left(F_{in}\right), \mathrm{Shift}_h^{+1}\left(F_{in}\right) \\ &\mathrm{Shift}_v^{-1}\left(F_{in}\right), \mathrm{Shift}_v^{+1}\left(F_{in}\right)\Big] \odot \mathrm{unsqueeze}\left(V\right) \end{aligned} \tag{28}$$

where $\mathrm{Shift}_h^{\pm 1}(F_{in})$ represents the operation of circularly shifting F_{in} by one pixel to the left (-1) or right $(+1)$ along the horizontal direction, while $\mathrm{Shift}_v^{\pm 1}(F_{in})$ denotes circularly shifting F_{in} by one pixel upward (-1) or downward $(+1)$ along the vertical direction, Stack$(\cdot)$ operation denotes stacking the features shifted in four directions along a newly created dimension to form a unified tensor structure. Subsequently, the relationships between the nodes V and edges E are refined and updated based on the constructed graph. A linear transformation is first applied to project the edge set E back to the same shape as the node set, denoted as $\mathrm{RT}(E) \in \mathbb{R}^{C_s \times H_s \times W_s}$. Then, the node features are updated according to the structural relationships among the nodes, resulting in intermediate node features $\tilde{V} \in \mathbb{R}^{C_s/2 \times H_s \times W_s}$. This process can be formulated as

$$\tilde{V} = \mathrm{T}\left\{\mathrm{Cat}\left[V, \mathrm{RT}\left(E\right)\right]\right\} \tag{29}$$

where RT$(\cdot)$ denotes the operation that reshapes the feature tensor back to its original dimensional space followed by a linear transformation, while T$(\cdot)$ represents the linear mapping performed by a fully connected layer.

Next, the initial node feature V is replicated four times to construct a directional dimension corresponding to the top, bottom, left, and right directions. The resulting feature is concatenated with the edge feature E along the channel dimension, followed by a linear transformation to generate the updated edge representation $\tilde{E} \in \mathbb{R}^{C_s/2 \times H_s \times W_s}$. This process can be formulated as

$$\tilde{E} = \mathrm{RT}\left(\mathrm{Cat}\left(\mathrm{Stack}\left(V^{\uparrow}, V^{\downarrow}, V^{\leftarrow}, V^{\rightarrow}\right), E\right)\right). \tag{30}$$

Subsequently, the intermediate node features $\tilde{V}$ and the updated edge features $\tilde{E}$ are fused through element-wise multiplication. The fused features are then concatenated with F_d and passed through a linear transformation to generate the final fused representation F_t. This process can be formulated as

$$F_t = \mathrm{T}\left(\mathrm{Cat}\left(\tilde{V} \odot \tilde{E}, F_d\right)\right). \tag{31}$$

Moreover, we introduce the subsequent bidirectional gating mechanism. In this mechanism, the fused feature F_t is employed as a supervisory signal to selectively filter the information flows of F_l and S_i'. Meanwhile, learnable parameters are utilized to adaptively assign the relative importance of shallow and deep features, enabling a weighted fusion. This process can be formulated as

$$\mathrm{Z}_l = \mathrm{Reshape}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{GAP}\left(F_l\right)\right)\right) \tag{32}$$

$$\mathrm{Z}_g = \mathrm{Reshape}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{GAP}\left(S_i'\right)\right)\right) \tag{33}$$

$$\mathrm{Z}_t = \mathrm{Reshape}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{GAP}\left(F_t\right)\right)\right) \tag{34}$$

$$\mathrm{Z}_{\mathrm{fuse}} = \beta\left(\mathrm{Z}_t \otimes \mathrm{Z}_l\right) + \left(1-\beta\right)\left(\mathrm{Z}_t \otimes \mathrm{Z}_g\right) \tag{35}$$

where $\otimes$ denotes matrix multiplication, and β is a learnable parameter. The resulting feature is further processed using a SoftMax operation, which modulates F_t. A residual connection is also employed to ensure stable information flow. This process can be formulated as

$$F_{\mathrm{dyn}} = \mathrm{Softmax}\left(\mathrm{Z}_{\mathrm{fuse}}\right) \otimes F_t \tag{36}$$

$$F_{\mathrm{out}} = \mathrm{CBR}\left(F_{\mathrm{dyn}}\right) + F_t \tag{37}$$

where Softmax$(\cdot)$ is a commonly used activation function that performs "normalization + probability conversion" on the input vector, enabling the model to produce outputs with well-defined probabilistic interpretation. CBR$(\cdot)$ consists of a 3×3 convolutional layer with subsequent batch normalization and ReLU activation. With this, the bidirectional optimization channels integrating F_t with both shallow and deep features are successfully constructed.

In summary, this framework leverages the SF-DAM enhanced features to introduce a structure-aware graph modeling mechanism, which strengthens the representation of spatial topological relationships and edge details. Coupled with the bidirectional gating mechanism, it enables efficient collaboration and dynamic fusion across multiple branches. Consequently, it alleviates information loss caused by semantic feature discrepancies and scale mismatches, thereby improving the accuracy and robustness of CD in complex scenarios.

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D. Loss Function

In CD tasks, a notable imbalance exists between changed and unchanged regions, with changed areas being much smaller. This imbalance often impairs the model's capability to accurately identify changes. To address this, we use a combined loss of BCE and Dice loss [44], with the BCE term defined as

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[gt_i \log\left(\hat{pr}_i\right) + (1-gt_i)\log\left(1-\hat{pr}_i\right)\right] \quad (38)$$

where N denotes the total number of pixels, $gt_i \in \{0,1\}$ represents the ground-truth label of the ith pixel in the binary form, and $\hat{pr}_i \in [0,1]$ indicates the predicted probability of change occurrence at the ith pixel location. The dice loss is mathematically formulated as

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2\sum_{i=1}^{N} gt_i \hat{pr}_i}{\sum_{i=1}^{N} gt_i^2 + \sum_{i=1}^{N} \hat{pr}_i^2} \quad (39)$$

where the dice loss effectively addresses the class imbalance problem by measuring the spatial overlap between the predicted region and the ground-truth region. It is particularly well-suited for CD tasks where the changed regions are usually sparse. The hybrid loss is formulated as

$$\mathcal{L}_{\mathrm{loss}} = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}}. \quad (40)$$

By combining BCE and Dice losses, the model achieves complementary pixel-level accuracy and regional-level consistency, thereby alleviating the sample imbalance problem.

IV. Experiments and Results Analysis

We evaluate the practical performance of SFGBO on three widely used public datasets. This section begins with an overview of the datasets and comparison methods, followed by the evaluation metrics and experimental setup, and concludes with a summary and analysis of the results. The alternative SF-DAM configurations evaluated in the ablation study are illustrated in Fig. 4.

A. Datasets

BCDD [1]: The dataset focuses on postearthquake building CD and contains two aerial images captured in 2012 and 2016, spanning roughly 20.5 km^2 with a spatial resolution of 0.2 m/pixel. It comprises 12 796 damaged buildings and 16 077 reconstructed buildings. The original images are divided into nonoverlapping patches of size 256×256 pixels and randomly split into 5895 training pairs, 794 validation pairs, and 745 testing pairs.

LEVIR-CD [45]: This dataset is a large-scale benchmark for building CD using remote sensing images. It contains 637 pairs of ultrahigh-resolution Google Earth patches at 0.5 m/pixel, each measuring 1024×1024 pixels. The images, collected from 20 regions in Texas, USA, include various building types such as warehouses, villas, garages, and high-rise apartments, with acquisition dates spanning 2002–2018 to capture temporal changes. For training and evaluation, the images are divided into nonoverlapping 256×256 patches, resulting in 7120 training pairs, 1024 validation pairs, and 2048 testing pairs.

SYSU-CD [46]: The SYSU-CD dataset, released by Sun Yat-sen University, is a large-scale, high-resolution RSCD benchmark. Compared with previous mainstream CD datasets, it offers higher image resolution, greater diversity of change types, and a larger data volume. The dataset comprises 20 000 aerial image pairs from the Hong Kong region, China, collected between 2007 and 2014 at a spatial resolution of 0.5 m/pixel. Each image is sized 256×256 pixels and paired with a corresponding binary change mask. The dataset captures various urban changes, including road expansion, building construction and demolition, and site reconstruction. Following the standard split, 12 000 pairs are used for training, 4000 for validation, and 4000 for testing.

Table I summarizes the detailed characteristics of the three datasets, including spatial resolution, patch size, data splits for training, validation, and testing, as well as the total number of pixels.

B. Comparison Methods

We evaluate the proposed SFGBO against existing RSCD methods that represent the current state-of-the-art in CD.

1) *FC-Siam-Conc [18]:* This method captures change information by concatenating and fusing feature maps, while preserving spatial integrity through the integration of shallow features.
2) *FC-Siam-Diff [18]:* This method employs differential computation to directly compare the feature maps of bitemporal images, enhancing the detection of changed regions while preserving spatial detail fidelity.
3) *ChangeFormer [22]:* This method is a CD framework that leverages Transformer and multilayer perceptron (MLP) architectures to model long-range dependencies, enabling effective performance on RSCD tasks.
4) *BIT-CD [17]:* This method compresses bitemporal features into semantic tokens using a semantic encoder, which are then processed by a Transformer to capture spatiotemporal relationships and achieve precise CD.
5) *DMINet [47]:* This method utilizes the JointAtt module, integrating self- and cross-attention to mitigate distractions and concentrate on regions of change. Detection performance is further improved through a multilevel aggregation strategy.
6) *CF-GCN [48]:* This method captures long-range dependencies and feature interactions in dual-temporal images through coordinate-space convolution, while simultaneously suppressing interference from "non-semantic variations" via boundary-aware and knowledge-retrieval modules.
7) *ELGC-Net [23]:* This method introduces the ELGCA module, which combines local-context depthwise convolutions with global-context pool-and-transpose attention to achieve efficient multiscale feature aggregation.
8) *SEIFNet [20]:* This method employs a dual-branch structure with subtraction and concatenation, combining coordinate attention and convolutions to capture spatiotemporal information while suppressing noise.

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TABLE I
INFORMATION ON THE MAIN PARAMETERS OF THREE PUBLIC DATASETS

Datasets	Spatial Resolution	Size/Image	Number of Pixels			Number of Dataset		
			Changed	Unchanged	Ratio	Train	Val	Test
BCDD	0.2 m/pixel	256×256	21,352,815	477,759,663	1:22.37	5895	794	745
LEVIR-CD	0.5 m/pixel	256×256	30,913,975	637,028,937	1:20.61	7120	1024	2048
SYSU-CD	0.5 m/pixel	256×256	286,092,024	1,024,627,976	1:3.58	12000	4000	4000

9) *ChangeRD [49]:* This method mitigates the impact of registration accuracy on CD by spatially transforming intermediate-layer features in the Siamese network and alleviates spectral inconsistency via the AgCDC module.
10) *RHighNet [50]:* This method captures the dependencies of bimodal temporal images in spatial and channel dimensions by constructing TRAM and introduces HOSTIM to simulate complex high-order feature interactions. Through a multistage loss optimization strategy, it enhances the model's representation capability in complex backgrounds.
11) *STRobustNet [51]:* This method generates spatiotemporally robust representations for land-use categories through an RRG module, transforming bitemporal features into consistent class-level representations. This alleviates spatiotemporal inconsistencies and enables efficient inference while maintaining accuracy.
12) *CDMamba [52]:* This method introduces the SRCM module, which integrates Mamba's global feature extraction capability with the local feature extraction strength of convolutions, enabling more comprehensive image representations.

C. Assessment Metrics

We benchmark the proposed CD framework with six standard metrics—recall (Rec.), precision (Pre.), $F1$-score ($F1$), intersection over union (IoU), overall accuracy (OA), and Cohen's Kappa (κ)—whose formulations are summarized below

$$Rec. = \frac{TP}{TP + FN} \tag{41}$$

$$Pre. = \frac{TP}{TP + FP} \tag{42}$$

$$F1 = 2 \cdot \frac{Pre. \times Rec.}{Pre. + Rec.} \tag{43}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{44}$$

$$OA = \frac{TP + TN}{TP + FN + TN + FP} \tag{45}$$

$$p_e = \frac{(TN + FN) \cdot (TN + FP) + (FP + TP) \cdot (FN + TP)}{(TP + FP + TN + FN)^2} \tag{46}$$

$$\kappa = \frac{OA - p_e}{1 - p_e} \tag{47}$$

where TP is the count of changed pixels correctly classified as change. FP denotes the unchanged pixels misclassified as change. TN indicates unchanged pixels correctly labeled as no-change, and FN corresponds to changed pixels erroneously assigned to the no-change class.

D. Implementation Details

SFGBO is built on the PyTorch framework, with experiments performed on an NVIDIA A40 GPU. To enhance training data variability and reduce overfitting, input images are subjected to standard augmentations such as random scaling, flipping, and cropping. Optimization is carried out using Adam [53] with momentum 0.9, weight decay $1e^{-4}$, and $\beta_1 = 0.9, \beta_2 = 0.99$. The learning rate is initially set to $2e^{-4}$ and adjusted according to the polynomial schedule in [54]. Models are trained for 100 epochs per dataset with a batch size of 16, and the checkpoint yielding the highest $F1$ score on the validation set is selected for evaluation. All comparative methods are re-implemented using their official code and default settings on the same hardware to ensure fairness.

E. Comparative Experiments

To evaluate the performance of SFGBO, we perform a quantitative comparison against several state-of-the-art methods on three benchmark datasets: BCDD, LEVIR-CD, and SYSU-CD. The numerical results are presented in Tables II and III, while Table IV provides a summary of computational costs and model parameters.

Results on BCDD: As illustrated in Table II, our method achieves Kappa, OA, IoU, and $F1$ scores of 94.99%, 99.56%, 90.88%, and 95.22%, respectively. In comparison, the second-best method, CDMamba, reaches 90.77%, 99.21%, 83.79%, and 91.18% on these metrics. Specifically, our approach surpasses CDMamba by +4.22%, +0.35%, +7.09%, and +4.04% in terms of Kappa, OA, IoU, and $F1$, respectively. These results indicate that our method outperforms existing state-of-the-art approaches on the BCDD dataset.

The visualization results are illustrated in Fig. 5, which illustrates typical examples from the BCDD dataset. Specifically, in the large-scale building change scenarios, illustrated in Fig. 5(a) and (b), most competing methods tend to suffer from boundary blurring or large-area omissions. In contrast, our approach precisely delineates building boundaries and captures more complete change contours, while effectively suppressing shadow interference to produce clearer edge information. In the complex boundary scenarios of samples, illustrated in Fig. 5(c) and (d), other methods exhibit varying degrees of false positives and missed detections in the details, whereas our method significantly enhances the representation of fine-grained structures. Finally, in the small-scale building change cases of samples, as shown in Fig. 5(e) and (f), most methods fail to detect small objects or produce distorted contours. Our approach, however, accurately captures small targets with

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TABLE II

QUANTITATIVE COMPARISON OF VARIOUS METHODS ON BCDD AND LEVIR-CD. THE BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Model	BCDD						LEVIR-CD					
	F1 (%)	κ (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)	F1 (%)	κ (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)
FC-Siam-Diff	82.70	81.92	89.10	77.16	70.51	98.49	86.22	85.49	87.57	84.91	75.78	98.62
FC-Siam-Conc	78.93	77.97	84.47	74.07	65.19	98.15	85.46	84.67	85.40	85.51	74.61	98.52
ChangeFormer	90.02	89.55	93.32	86.95	81.85	99.10	90.01	89.48	91.92	88.17	81.83	99.00
BIT-CD	84.27	83.52	85.74	82.86	72.82	98.56	89.95	89.43	**93.19**	86.92	81.73	99.01
DMINet	86.61	85.94	84.59	88.72	76.38	98.72	90.69	90.21	93.19	88.32	82.97	99.08
CF-GCN	90.32	89.86	93.72	87.15	82.34	99.13	90.39	89.88	91.74	89.07	82.46	99.04
ELGC-Net	89.12	88.62	93.64	85.03	80.38	99.03	89.95	89.42	91.42	88.52	81.73	98.99
SEIFNet	87.47	86.86	87.39	87.55	77.73	98.83	90.25	89.73	91.89	88.66	82.23	99.02
ChangeRD	90.09	89.63	94.20	86.32	81.97	99.11	89.63	89.08	91.53	87.80	81.20	98.97
RHighNet	90.30	89.85	94.65	86.34	82.32	99.13	90.91	90.43	92.53	89.34	83.33	99.09
STRobustNet	89.48	88.99	92.92	86.29	80.97	99.05	90.71	90.22	92.15	89.32	83.01	99.07
CDMamba	91.18	90.77	94.77	87.86	83.79	99.21	90.29	89.78	91.82	88.82	82.31	99.03
Ours	**95.22**	**94.99**	**96.37**	**94.11**	**90.88**	**99.56**	**91.80**	**91.37**	92.87	**90.76**	**84.85**	**99.17**

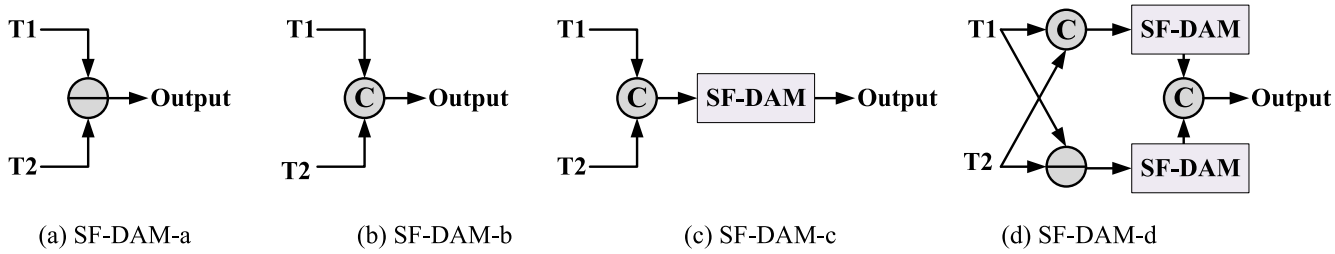


Fig. 4. (a) Absolute subtraction. (b) Concatenation. (c) Connection-based enhancement. (d) Connection- and difference-based enhancement.

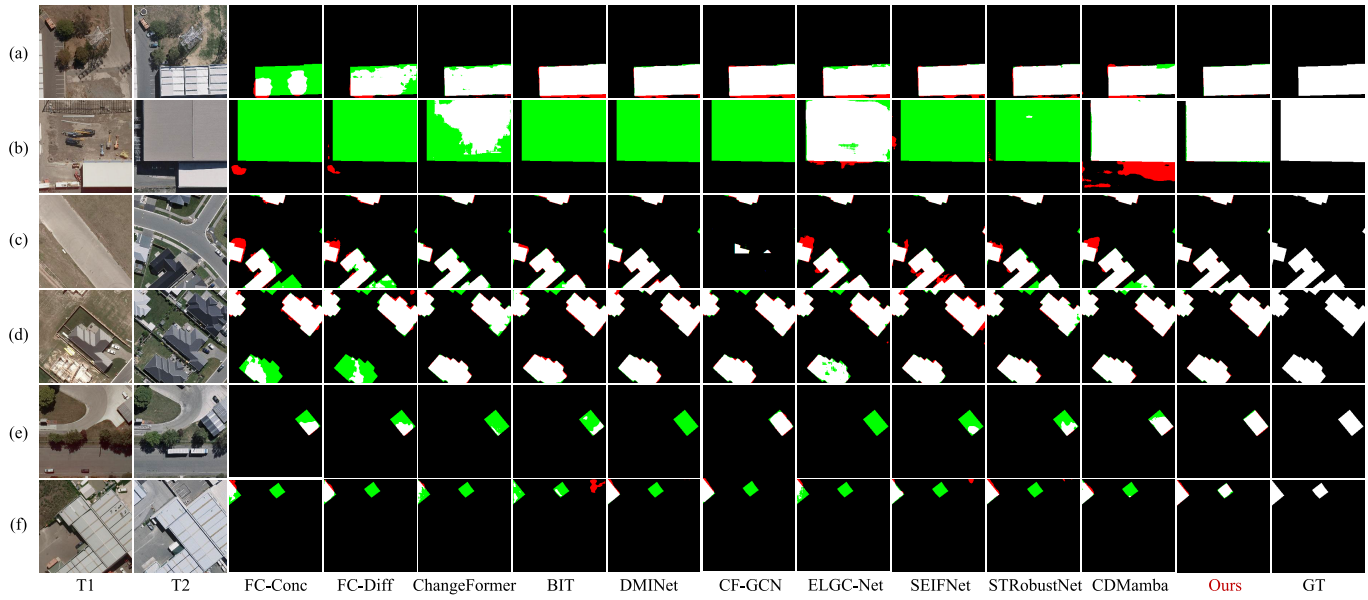


Fig. 5. Visualization of various models on the BCDD dataset. (a)–(f) Show representative samples. GT denotes the ground truth. Changed objects are highlighted in white, while unchanged objects are shown in black. False positives are marked in red, and false negatives are marked in green.

well-preserved structural consistency and almost no noise interference.

Results on LEVIR-CD: As illustrated in Table II, our method achieves 91.37%, 99.17%, 84.85%, and 91.80% in terms of Kappa, OA, IoU, and $F1$, respectively. In comparison,

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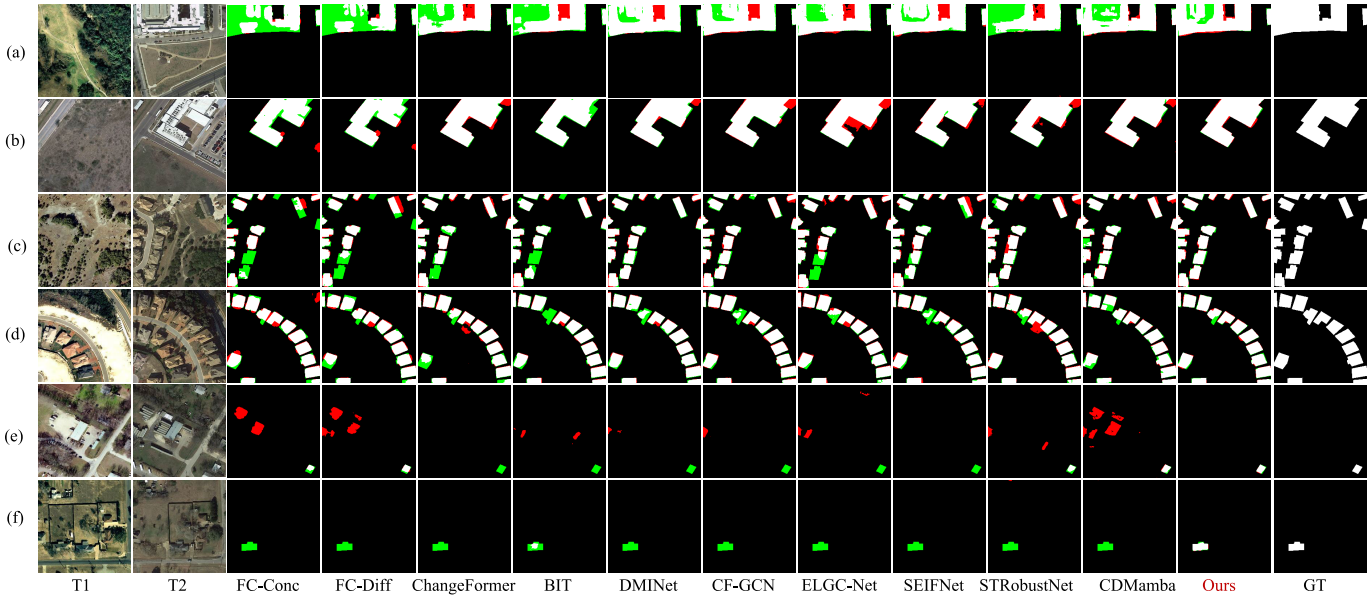


Fig. 6. Visualization of various models on the LEVIR-CD dataset. (a)–(f) Show representative samples. GT denotes the ground truth. Changed objects are highlighted in white, while unchanged objects are shown in black. False positives are marked in red, and false negatives are marked in green.

TABLE III

QUANTITATIVE COMPARISON OF VARIOUS METHODS ON SYSU-CD. THE BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Model	SYSU-CD					
	F1 (%)	κ (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)
FC-Siam-Diff	67.86	59.81	79.88	58.99	51.35	86.82
FC-Siam-Conc	79.54	73.27	79.90	79.19	66.04	90.39
ChangeFormer	78.20	71.43	77.81	78.60	64.21	89.67
BIT-CD	77.44	70.63	78.79	76.12	63.18	89.54
DMINet	81.00	75.39	83.84	78.35	68.07	91.33
CF-GCN	78.98	73.08	84.96	73.79	65.26	90.74
ELGC-Net	78.82	72.74	83.28	74.81	65.05	90.52
SEIFNet	<u>82.08</u>	<u>76.57</u>	82.30	**81.87**	<u>69.61</u>	<u>91.57</u>
RHighNet	80.08	74.41	<u>85.24</u>	75.51	66.78	91.14
STRobustNet	79.62	73.32	79.60	79.63	66.13	90.38
CDMamba	78.79	72.49	81.12	76.59	65.00	90.28
Ours	**82.72**	**77.63**	**85.64**	<u>80.00</u>	**70.54**	**92.12**

TABLE IV

PARAMETERS, COMPUTATIONAL COMPLEXITY, AND CORRESPONDING $F1$-SCORES AND κ COEFFICIENTS OF THE COMPARISON METHODS ON THE LEVIR DATASET

Model	Params (M)	FLOPs (G)	F1 (%)	κ (%)
FC-Siam-Diff	1.35	4.72	86.22	85.49
FC-Siam-Conc	1.55	5.30	85.46	84.67
ChangeFormer	41.03	202.79	90.01	89.48
BIT-CD	3.50	10.63	89.95	89.43
DMINet	6.24	14.55	90.69	90.21
CF-GCN	13.58	44.08	90.39	89.88
ELGC-Net	10.57	187.99	89.95	89.42
SEIFNet	27.91	8.37	90.25	89.73
ChangeRD	29.47	18.27	89.63	89.08
RHighNet	96.81	66.51	90.91	90.43
STRobustNet	13.73	20.99	90.71	90.22
CDMamba	11.91	49.26	90.29	89.78
Ours	11.54	36.54	**91.80**	**91.37**

the second-best method, RHighNet, obtains 90.43%, 99.09%, 83.33%, and 90.91% on the same metrics. Specifically, our approach outperforms RHighNet by +0.94%, +0.08%, +1.52%, and +0.89% on Kappa, OA, IoU, and $F1$, respectively. These results indicate that our method outperforms existing state-of-the-art approaches on the LEVIR-CD dataset.

The visualization results are illustrated in Fig. 6, which illustrates typical examples from the LEVIR-CD dataset, highlighting diverse building change patterns. Specifically, in the large-scale building scenes of samples in Fig. 6(a) and (b), our network exhibits fewer missed detections and captures more complete building outlines compared to other competing methods. In the densely changed scenes of samples in Fig. 6(c) and (d), our model demonstrates higher sensitivity and robustness to interference, effectively detecting small-scale edge changes while mitigating background disturbances caused by illumination and weather variations, highlighting its strong change-focusing capability. For the small building cases, in Fig. 6(e) and (f), other methods struggle with detecting minor changes, resulting in missed and false detections. In contrast, our network performs well and produces the most visually accurate results.

Results on SYSU-CD: As illustrated in Table III, our method achieves Kappa, OA, IoU, and $F1$ scores of 77.63%, 92.12%, 70.54%, and 82.72%, respectively. In comparison, the second-best method, SEIFNet, obtains 76.57%, 91.57%, 69.61%, and 82.08% on these metrics. Specifically, our approach

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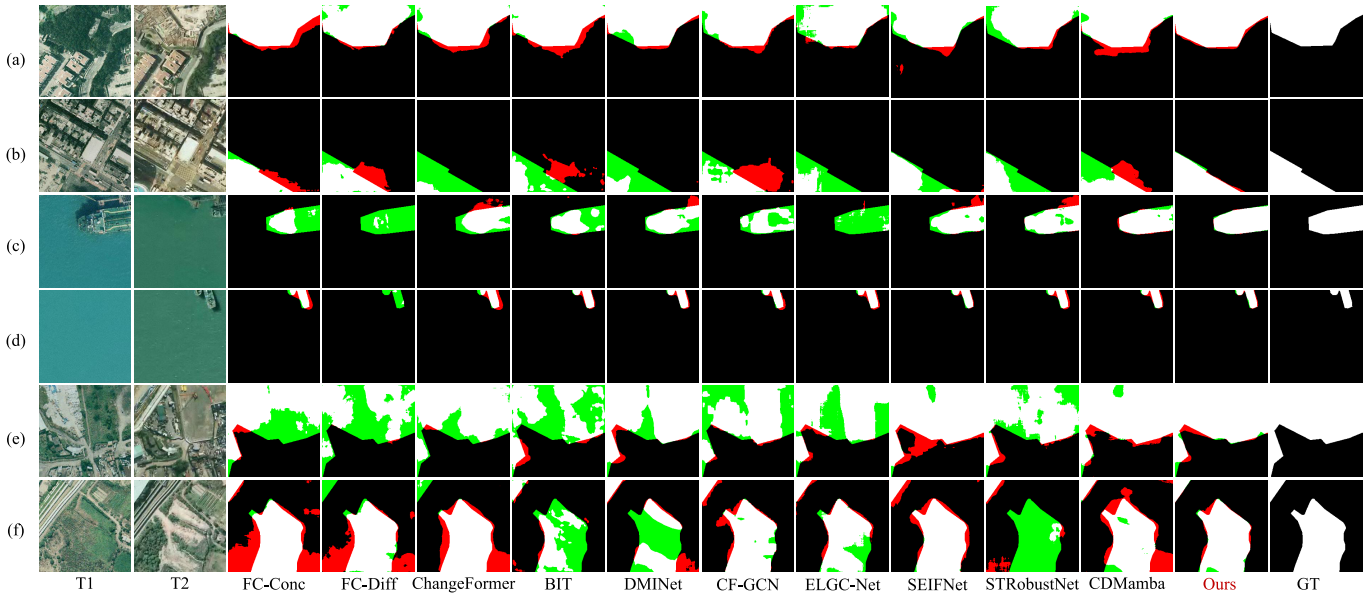


Fig. 7. Visualization of various models on the SYSU-CD dataset. (a)–(f) Show representative samples. GT denotes the ground truth. Changed objects are highlighted in white, while unchanged objects are shown in black. False positives are marked in red, and false negatives are marked in green.

outperforms SEIFNet by +1.06%, +0.55%, +0.93%, and +0.64% in terms of Kappa, OA, IoU, and $F1$, respectively. These results clearly demonstrate that our method exhibits stronger competitiveness on the SYSU-CD dataset compared with other approaches.

The visualization results are illustrated in Fig. 7, which illustrates typical examples from the SYSU-CD dataset that feature more challenging and diverse change scenarios. Specifically, in the urban redevelopment scenes of samples, as shown in Fig. 7(a) and (b), most methods generate coarse building contours, whereas our method produces more reliable and accurate boundary delineations. In the water-related scenes of samples, as shown in Fig. 7(c) and (d), most methods struggle with blurred or inaccurate boundaries around ship shadows. By comparison, our method produces cleaner black backgrounds for unchanged water regions, while the detected changes closely match the true target shapes. Finally, in the land development and grassland degradation scenes, as shown in Fig. 7(e) and (f), many methods suffer from missing small change regions and misclassifying unchanged areas as changes. In contrast, our approach generates more coherent and complete change maps, achieving a higher correspondence with the ground-truth changes and demonstrating superior performance in both the *integrity* and *precision* of complex land-use CD.

In summary, compared to other approaches, the proposed SFGBO achieves superior overall performance, with the lowest false positive and false negative rates, underscoring its applicability and competitive advantage in the field of RSCD.

F. Ablation Studies

To evaluate the contributions of the components and configurations of SFGBO, we conduct ablation experiments on two benchmark datasets, with the results reported in Table V. In Table V, #00 denotes the baseline model. This model retains the same backbone, decoder structure, and loss function as the full model, with only the SF-DAM and GBDM modules removed.

1) Impact of Different Backbones and Backbone-Matched Performance Analysis: To analyze the impact of backbone networks on the proposed framework, we conduct comparative experiments under identical settings using seven different backbones, including MobileNetV2, ResNet18, ResNet34, ResNet50, Swin Transformers, DINOv3 (ConvNeXt), and VGG16, on three public datasets.

As shown in Tables VI and VII, the optimal backbone varies across datasets. Specifically, VGG16 achieves the best performance on LEVIR-CD and BCDD, while ResNet34 obtains the best results on SYSU-CD. This phenomenon indicates that different datasets may favor different backbone architectures due to variations in scene complexity, object categories, and change patterns. LEVIR-CD and BCDD are mainly dominated by building-related changes with relatively regular geometric structures and clear boundaries, which can benefit from the strong local feature extraction capability of VGG16. In contrast, SYSU-CD contains more diverse and heterogeneous changes, such as roads, vegetation, water-related areas, and mixed urban transitions, requiring stronger feature generalization and robustness. Therefore, residual backbones such as ResNet34 show better adaptability on this dataset. It is worth noting that advanced backbones such as Swin Transformers and DINOv3 (ConvNeXt), despite their strong feature representation capabilities, do not outperform VGG16 in our experiments. One possible explanation is that these models, due to their higher complexity, may be more sensitive to limited training data and prone to capturing irrelevant patterns, which can affect generalization. In comparison, the relatively simple architecture of VGG16 focuses more on

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TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT CONFIGURATIONS OF THE PROPOSED METHOD ON THE LEVIR-CD AND BCDD DATASETS

No.	Variants	LEVIR-CD					BCDD				
		F1 (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)	F1 (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)
#00	Baseline	90.29	91.33	89.27	82.30	99.02	93.89	95.29	92.52	88.48	99.44
#01	SFGBO	91.80	92.87	90.76	84.85	99.17	95.22	96.37	94.11	90.88	99.56
Spatial-Frequency Synergistic Dynamic Attention Module(SF-DAM)											
#02	SF-DAM-a	91.47	92.90	90.08	84.28	99.14	94.82	95.92	93.76	90.16	99.52
#03	SF-DAM-b	91.69	92.95	90.47	84.66	99.16	94.43	96.13	92.79	89.45	99.49
#04	SF-DAM-c	91.70	93.61	89.86	84.67	99.17	94.42	96.80	92.16	89.43	99.49
#05	SF-DAM-d	91.68	92.97	90.43	84.64	99.16	95.07	96.02	94.13	90.60	99.54
#06	w/o Fre	91.52	94.13	89.05	84.36	99.16	94.80	95.98	93.64	90.11	99.52
Graph-Based Bidirectional Dynamic Modeling Module(GBDM)											
#07	Add	90.86	91.61	90.11	83.24	99.08	94.36	96.17	92.43	89.32	99.48
#08	Concat	90.97	91.83	90.13	83.44	99.09	94.41	96.14	92.74	89.41	99.49
#09	w/o BOM	91.51	92.24	90.78	84.34	99.14	94.68	96.63	92.82	89.91	99.51
Loss Functions											
#10	BCE	91.16	91.47	90.86	83.76	99.10	94.70	96.91	92.59	89.93	99.52
#11	Dice	91.55	92.84	90.28	84.41	99.15	94.58	96.45	92.78	89.71	99.50

TABLE VI
PERFORMANCE COMPARISON ACROSS TWO DATASETS USING DIFFERENT BACKBONES. FPS IS MEASURED UNDER A UNIFIED PROTOCOL WITH BATCH SIZE 1, INPUT RESOLUTION 256 × 256, 100 WARM-UP ITERATIONS, 2000 INFERENCE ITERATIONS, AND CUDA SYNCHRONIZATION ON THE SAME GPU

Model	Backbone	Efficiency		LEVIR-CD					BCDD				
		FPS	Params(M)	F1 (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)	F1 (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)
SFGBO	MobileNetV2	66.70	8.28	91.14	90.32	91.96	83.72	99.11	94.62	96.38	92.93	89.79	99.51
SFGBO	ResNet18	92.42	8.95	91.09	89.94	92.28	83.64	99.10	95.00	96.56	93.49	90.48	99.54
SFGBO	ResNet34	77.50	14.34	91.36	90.27	**92.47**	84.09	99.13	94.36	**96.86**	91.99	89.33	99.49
SFGBO	ResNet50	68.82	15.23	91.60	90.74	92.47	84.49	99.15	94.95	96.61	93.35	90.39	99.54
SFGBO	VGG16	105.23	11.54	**91.80**	**92.87**	90.76	**84.85**	**99.17**	**95.22**	96.37	**94.11**	**90.88**	**99.56**
SFGBO	Swin Transformers	41.42	34.01	91.27	91.69	90.85	83.94	99.11	94.43	96.40	92.54	89.45	99.49
SFGBO	DINOv3 (ConvNeXt)	58.98	34.31	91.04	92.18	89.92	83.55	99.10	93.23	95.65	90.93	87.32	99.38

TABLE VII
PERFORMANCE COMPARISON ACROSS TWO DATASETS USING DIFFERENT BACKBONES

Backbone	FLOPs	SYSU-CD				
		F1 (%)	Pre. (%)	Rec. (%)	IoU (%)	OA (%)
MobileNetV2	5.31	83.73	84.50	82.98	72.02	92.40
ResNet18	8.25	83.33	83.49	83.18	71.43	92.15
ResNet34	12.49	**83.82**	84.49	83.16	**72.15**	**92.43**
ResNet50	13.84	83.61	85.40	81.90	71.84	92.43
VGG16	36.54	82.72	**85.64**	80.00	70.54	92.12
Swin Transformers	15.52	82.88	80.51	**85.39**	70.77	91.68
DINOv3 (ConvNeXt)	12.94	82.14	84.34	80.05	69.69	91.79

local structural variations, which are particularly important for building-centric CD tasks, thereby leading to more stable performance. In addition, the inference efficiency of different backbones is also an important factor to consider. It should be noted that inference speed (FPS) is not solely determined by parameter size or FLOPs. For example, although MobileNetV2 is lightweight, its extensive use of depthwise separable convolutions can lead to lower arithmetic intensity and less efficient GPU utilization, meaning that its practical FPS may not scale proportionally with its theoretical computational cost. Similarly, deeper architectures such as ResNet34 and ResNet50 introduce more convolutional layers, normalization, and residual operations, which increase memory access overhead and limit runtime efficiency. Swin Transformer and DINOv3 (ConvNeXt) also show relatively low FPS, possibly due to the additional overhead from attention-based token operations and more complex modern convolutional blocks, respectively. In contrast, VGG16 and ResNet18 involve fewer layers and simpler computational paths, resulting in lower memory access overhead and more efficient execution on GPUs, which leads to higher practical throughput.

Although the above results indicate that backbone selection affects absolute performance, they do not fully reveal whether the performance gains come from the proposed method

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TABLE VIII
PERFORMANCE GAINS OF SFGBO OVER THE BASELINE WITH DIFFERENT BACKBONES ON THE LEVIR-CD AND SYSU-CD DATASETS

Backbone	LEVIR-CD			SYSU-CD		
	F1/IoU (Baseline)	F1/IoU (SFGBO)	F1/IoU (Gain)	F1/IoU (Baseline)	F1/IoU (SFGBO)	F1/IoU (Gain)
MobileNetV2	88.91 / 80.03	91.14 / 83.72	+2.23 / +3.69	81.35 / 68.56	83.73 / 72.02	+2.38 / +3.46
ResNet18	89.79 / 81.47	91.09 / 83.64	+1.30 / +2.17	81.40 / 68.64	83.33 / 71.43	+1.93 / +2.79
ResNet34	90.22 / 82.18	91.36 / 84.09	+1.14 / +1.91	82.20 / 69.77	83.82 / 72.15	+1.62 / +2.38
ResNet50	90.56 / 82.75	91.60 / 84.49	+1.04 / +1.74	81.96 / 69.44	83.61 / 71.84	+1.65 / +2.40
VGG16	90.29 / 82.30	91.80 / 84.85	+1.51 / +2.55	79.96 / 66.62	82.72 / 70.54	+2.76 / +3.92
Swin Transformers	89.84 / 81.56	91.27 / 83.94	+1.43 / +2.38	81.83 / 69.25	82.88 / 70.77	+1.05 / +1.52
DINOv3 (ConvNeXt)	90.34 / 82.38	91.04 / 83.55	+0.70 / +1.17	80.21 / 66.96	82.14 / 69.69	+1.93 / +2.73

itself or from the backbone choice. To further address this issue, we conduct backbone-matched comparisons between the baseline and the full SFGBO under the same backbone settings. This controlled comparison isolates the contribution of the proposed modules from the influence of backbone architectures. For clarity, we present the backbone-matched results on LEVIR-CD and SYSU-CD, which are representative of structured and complex change scenarios, respectively. As shown in Table VIII, SFGBO consistently outperforms the corresponding baseline across all tested backbones on both LEVIR-CD and SYSU-CD datasets. Specifically, the $F1$-score improvements range from +0.70% to +2.23% on LEVIR-CD and from +1.05% to +2.76% on SYSU-CD. Similarly, the IoU improvements range from +1.17% to +3.69% on LEVIR-CD and from +1.52% to +3.92% on SYSU-CD. These consistent improvements demonstrate that the performance gains are mainly attributed to the proposed SF-DAM and GBDM modules rather than a specific backbone.

In summary, backbone selection influences the performance–efficiency trade-off of the proposed framework, and different datasets may benefit from different backbone architectures. Nevertheless, the backbone-matched results verify that SFGBO can consistently improve performance under different backbone choices, indicating that the proposed method has good robustness and general applicability.

2) Effectiveness of the SF-DAM: The proposed SFGBO integrates the SF-DAM module to collaboratively process multiscale features in both spatial and frequency domains, enhancing the discrimination between change regions and background while highlighting fine-grained textures of small targets. To examine the impact of this component, we performed experiments replacing it with a basic absolute difference operation, as illustrated in Fig. 4(a). This variant is referred to as SF-DAM-a (#02 in Table V), and the corresponding quantitative results are reported. The model's performance decreased on both the LEVIR-CD and BCDD datasets: on LEVIR-CD, $F1$ and IoU dropped to 91.47% and 84.28%, respectively, compared to 91.80% and 84.85% for the full model; on BCDD, $F1$ and IoU were 94.82% and 89.79%, also lower than 95.22% and 90.88% of the complete model. These results indicate that SF-DAM plays a positive role in enhancing the discriminative capability of change regions.

To further demonstrate the rationality of the current architecture, we implemented three alternative variants, as illustrated in Fig. 4(b)–(d). The results indicate that although these variants increase the feature processing intensity, their actual performance decreases to varying degrees. This suggests that higher model complexity does not necessarily lead to better effectiveness. Furthermore, we propose a variant model that removes only the frequency-domain processing while preserving the remaining spatial and attention components. As shown in Table V #06, this variant exhibits varying degrees of metric degradation across both datasets. This further validates the advantages and necessity of the current design in capturing spatial-frequency differences and performing effective feature aggregation.

3) Effectiveness of the GBDM: In the proposed SFGBO, the GBDM module is designed to efficiently leverage shallow-layer information. To validate its effectiveness, we conducted three sets of ablation experiments. When the GBDM module was removed and features were fused using simple addition (#07 in Table V) or concatenation (#08 in Table V), the model's $F1$ scores on LEVIR-CD dropped to 90.86% and 90.97%, respectively, both lower than 91.80% of the full model. Similarly, on BCDD, $F1$ scores decreased to 94.36% and 94.41%, significantly weaker than 95.22% of the complete model. Furthermore, when the bidirectional optimization mechanism within GBDM was removed (#09 in Table V), performance also declined: on LEVIR-CD, $F1$ and IoU were 91.51% and 84.34%, below 91.80% and 84.85% of the full model; on BCDD, $F1$ and IoU were 94.68% and 89.91%, lower than 95.22% and 90.88% of the complete model. These results indicate that simple addition or concatenation alone is insufficient to effectively model the complex relationships between shallow and deep features. In contrast, GBDM, leveraging a graph-guided bidirectional dynamic modeling mechanism, significantly enhances feature interaction and fusion, leading to superior performance in CD tasks.

4) Loss Function Discussion: We adopt a hybrid loss combining Dice and BCE, and its effectiveness is validated through experiments reported in Table V (#10, #11). The results indicate that, compared to using Dice or BCE alone, the hybrid loss performs better in addressing class imbalance. Specifically, BCE provides pixel-level supervision, while Dice emphasizes spatial overlap, making the hybrid loss particularly

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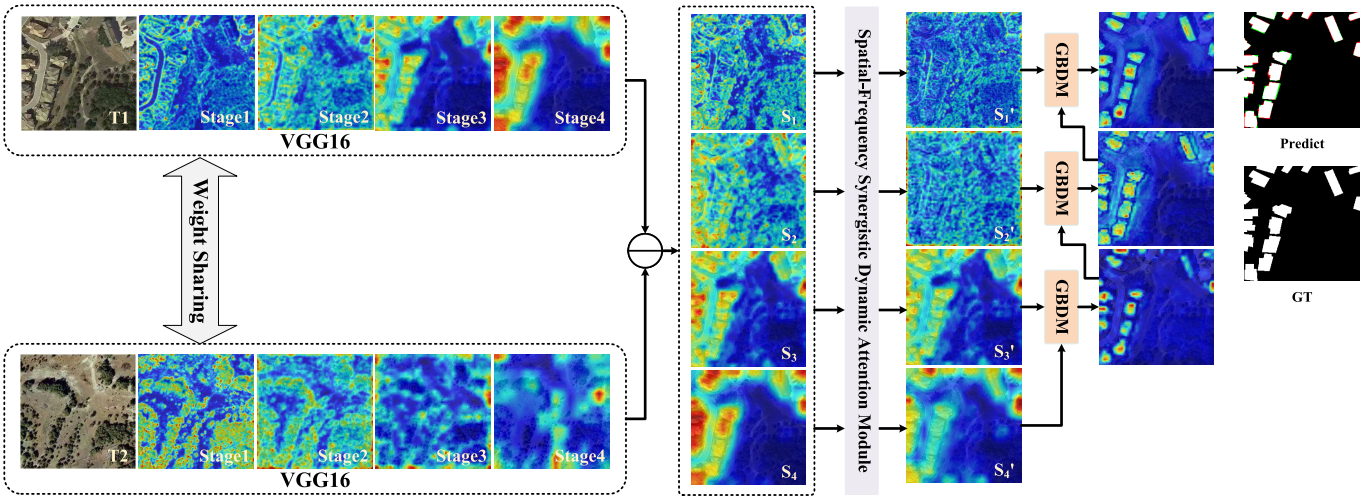


Fig. 8. Visualization of the model's decision-making process.

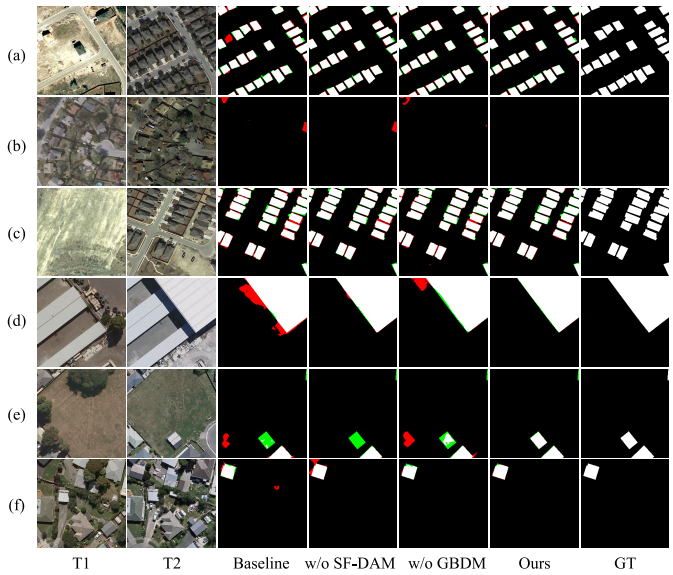


Fig. 9. Qualitative visualization results under different challenging scenarios. (a) and (b) Illustrate scenes affected by illumination variations. (c) and (d) Present cases with building shadow interference. (e) and (f) Show scenarios disturbed by tree occlusion.

advantageous in tasks where change regions are underrepresented.

In Fig. 8, to more effectively illustrate the SFGBO model, we visualize the feature maps at each stage. As can be observed from the visualization results, after processing with SF-DAM, noise caused by irrelevant changes, such as roads, is effectively suppressed, providing a solid foundation for subsequent information utilization. Moreover, with the application of GBDM, the model progressively produces change maps with improved visual quality during the decoding process. In addition, Fig. 9 presents qualitative visualization results under several representative scenarios, which further demonstrate the effectiveness of the proposed method in a more intuitive manner.

V. Conclusion

This article proposes a novel RSCD framework, termed SFGBO, which integrates a difference enhancement path and a graph-guided bidirectional dynamic modeling path to significantly improve detection accuracy and robustness in complex scenarios. In the encoding stage, the SF-DAM module jointly models spatial and frequency domains to emphasize fine-grained texture and structural information within subtle change regions. During the decoding stage, the GBDM module leverages deformable convolution and graph-based modeling to dynamically fuse shallow difference features with high-level semantic features. This design not only strengthens spatial topology and boundary representations but also mitigates information loss caused by semantic discrepancies and scale mismatches. Experiments on three mainstream CD datasets demonstrate that SFGBO achieves strong competitiveness. However, the proposed method still has certain limitations. For instance, the inference efficiency on large-scale high-resolution remote sensing images and the cross-domain generalization capability remain to be further improved. On the one hand, although the introduction of frequency-domain modeling helps enhance the discriminability of difference features, the learned frequency responses may still be influenced by variations in sensor characteristics, spatial resolution, and imaging conditions, which can reduce feature stability when transferring across sensors or geographic regions. On the other hand, the adopted graph-guided modeling mechanism relies on spatial topological relationships learned from the training data; when object scales, spatial layouts, or scene distributions change significantly across domains, the effectiveness of such graph-based interactions may be weakened. In addition, since different datasets vary in change types, object scales, and scene complexity, the most suitable backbone architecture may also differ, which introduces additional considerations for practical deployment. Future research will focus on developing a lightweight and general framework that maintains high efficiency while delivering strong performance across diverse scenarios, thereby improving the practicality and generalization capability of the model.

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